

Panel Data I

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Applied Quantitative Methods II
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- Cluster standard errors correctly in panel settings

Roadmap

What is Panel Data?

The Problem: Unobserved Heterogeneity

Fixed Effects

Two-Way Fixed Effects

Random Effects and the FE/RE Choice

Clustered Standard Errors

Wrap-up

Panel data: the basic structure

Unit i	Time t	y_{it}	x_{it}
1	2010	0.42	12.1
1	2011	0.51	13.0
1	2012	0.48	12.7
2	2010	0.61	9.4
2	2011	0.59	9.8
2	2012	0.64	10.2

- N units, each observed at T time points
- Data indexed (i, t) : unit i at time t

Panel data: examples

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- **Firms:** quarterly earnings reports for publicly traded companies
- **Running example:** US state-level presidential approval $\times$ years

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 - Units may differ in ways we cannot measure
 - Panel structure lets us “absorb” those differences

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Fixed Effects

Two-Way Fixed Effects

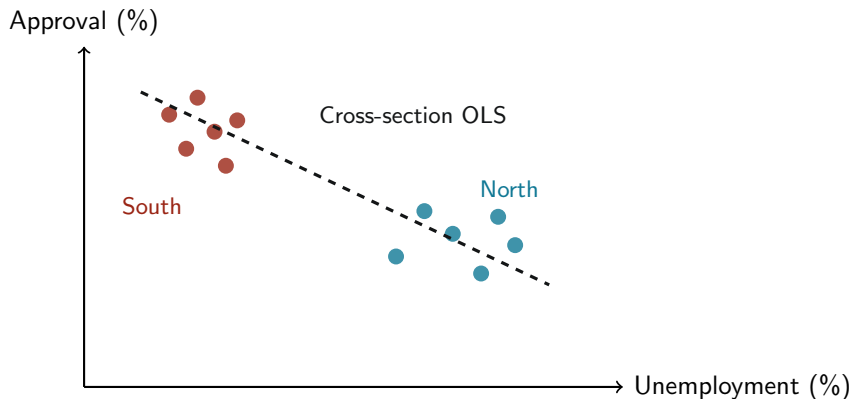
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Motivating example: presidential approval

Does unemployment drive down presidential approval?



The cross-sectional slope is negative.

Does that mean unemployment *causes* lower approval?

What else might explain this pattern?

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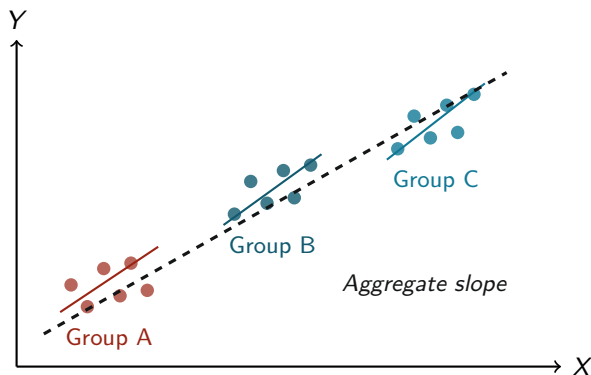
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- α_i = unit-specific intercept (the unobserved heterogeneity)
- Cross-section OLS ignores $\alpha_i \Rightarrow$ omitted variable bias

Simpson's paradox: the intuition



- Within each group: positive slope
- Cross-section OLS: also positive, but for the **wrong reason**
- The group-level differences dominate the estimate

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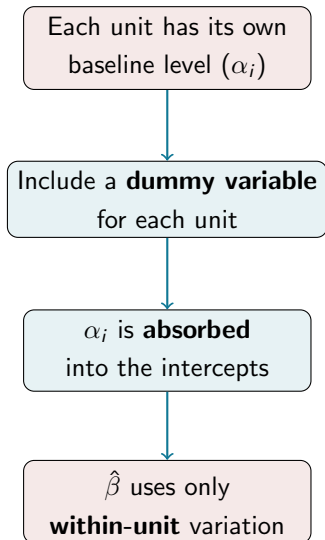
Two-Way Fixed Effects

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Fixed effects: the key idea



The within (demeaning) estimator

Starting from $y_{it} = \alpha_i + \beta x_{it} + \varepsilon_{it}$, subtract unit means:

$$\underbrace{y_{it} - \bar{y}_i}_{\tilde{y}_{it}} = \beta \underbrace{(x_{it} - \bar{x}_i)}_{\tilde{x}_{it}} + \underbrace{(\varepsilon_{it} - \bar{\varepsilon}_i)}_{\tilde{\varepsilon}_{it}}$$

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- Regressing $\tilde{y}_{it}$ on $\tilde{x}_{it}$ gives the FE estimator
- Uses only variation *within* each unit over time
- Works because α_i is constant: $\bar{\alpha}_i = \alpha_i$

FE = dummies for each unit

- Mathematically equivalent to including unit dummies:

$$y_{it} = \sum_{i=1}^N \alpha_i D_i + \beta x_{it} + \varepsilon_{it}$$

where $D_i = 1$ if observation belongs to unit i

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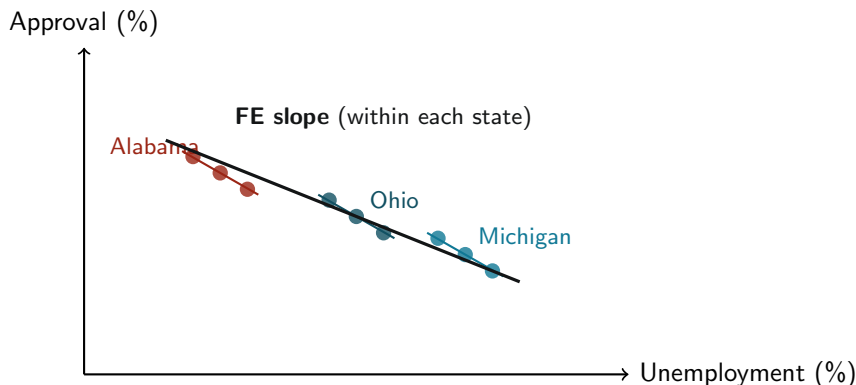
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 - Example: “South” dummy, gender, country of birth

Presidential approval: what FE does



- Each state has its own intercept; the slope is shared
- FE estimates: as *this state's* unemployment rises, its approval falls

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- Tables with `modelsummary()` work seamlessly with `feols` objects

Roadmap

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Wrap-up

FE controls for everything time-invariant.

But what if something happens in 2008 that affects *all* states simultaneously?

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- Solution: add **time fixed effects** γ_t

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- But more credible: fewer threats to identification

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Random effects: a different assumption

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- If this fails: RE is **biased**; FE is consistent

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- If this fails: RE is **biased**; FE is consistent
- Unlike FE: can estimate effects of **time-invariant** variables

FE vs. RE: the tradeoff

	Fixed Effects	Random Effects
Assumption	α_i correlated with X ?	$\eta_i \perp X$
Consistency	Always (if $T \rightarrow \infty$)	Only if $\eta_i \perp X$
Efficiency	Less efficient	More efficient
Time-invariant vars	Cannot estimate	Can estimate

- If you are unsure: **use FE**

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- RE requires an untestable assumption; FE does not

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- If $p < 0.05$: reject $H_0 \Rightarrow$ use FE

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- In R (plm package):

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- **Hausman test** (1978): formally test whether $\eta_i \perp x_{it}$
- H_0 : no correlation between unit effects and regressors (RE is consistent)
- H_1 : correlation exists (only FE is consistent)
- If $p < 0.05$: reject $H_0 \Rightarrow$ use FE
- If $p > 0.05$: fail to reject $\Rightarrow$ RE or FE both OK
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re_mod = plm(y ~ x, data = pdf, model = "random")
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Roadmap

What is Panel Data?

The Problem: Unobserved Heterogeneity

Fixed Effects

Two-Way Fixed Effects

Random Effects and the FE/RE Choice

Clustered Standard Errors

Wrap-up

Why panel data violates iid

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- Solution: **cluster standard errors by unit**

Clustering in practice

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- In `modelsummary()` tables:

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- Rule of thumb: cluster at the level of treatment assignment
 - If state gets the treatment, cluster by state
 - If individual gets the treatment, cluster by individual

Comparing specifications: a template

	(1) OLS	(2) Unit FE	(3) TWFE
Unemployment	-0.82*** (0.15)	-0.51** (0.18)	-0.43** (0.16)
State FE	No	Yes	Yes
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- Preferred specification: (3)

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 - Cannot estimate time-invariant variables
- **Two-way FE**: add time dummies to remove common shocks
- **FE vs. RE**: use FE when unit effects may correlate with X ; use Hausman test
- Always **cluster standard errors** by unit

For next session

- Complete Assignment 5
- Read the assigned paper using panel FE
- Next session: Panel Data II
 - Difference-in-Differences (DiD)
 - Event studies
 - Staggered treatment timing

Questions?